Understanding Customer Purchasing Patterns with RFM Analysis

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The Business Problem

- Customers differ in purchase timing, frequency, and spending
- A single engagement strategy may overlook those differences.
- Can purchasing patterns identify useful customer groups?
- What actions could a retailer consider for each group?

Data Source and Preparation

- UCI Online Retail dataset
- One UK retailer, December 2010 – December 2011
- Original: **541,909 transaction-line records**
- Retained: **392,692 records and 4,338 customers**
- Excluded missing customer IDs, cancellations, nonpositive quantities or prices, exact duplicates

Chen (2015)

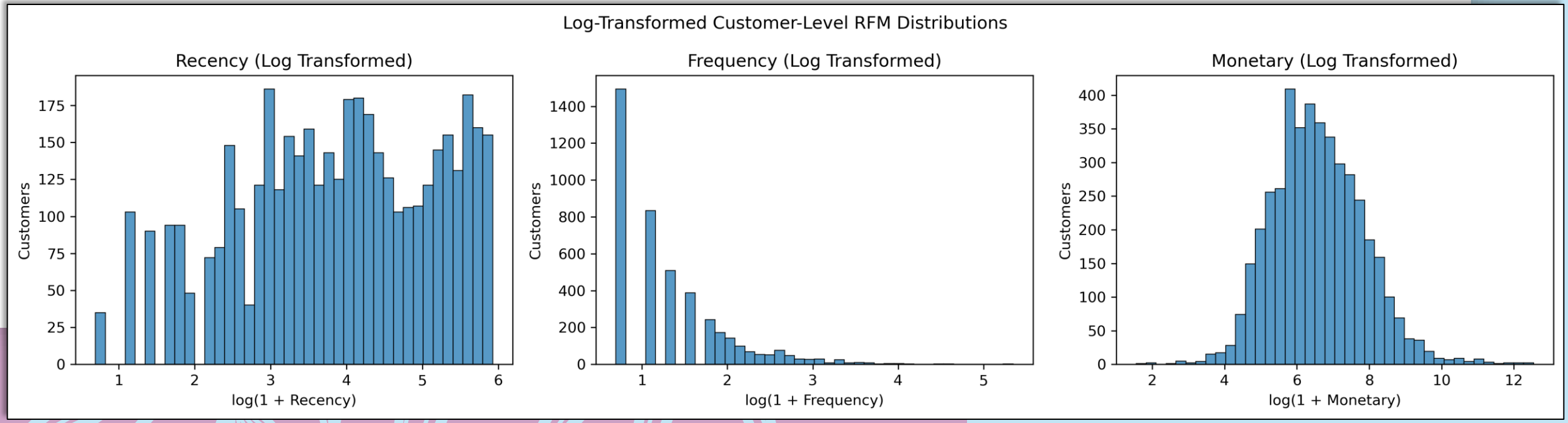
Measuring Purchasing Behavior

- Recency – Days since the most recent retained purchase
- Frequency – Number of distinct purchase invoices
- Monetary – Total retained positive purchasing value
- Recency reference date: December 10, 2011
- One record per customer

Hughes (1994)

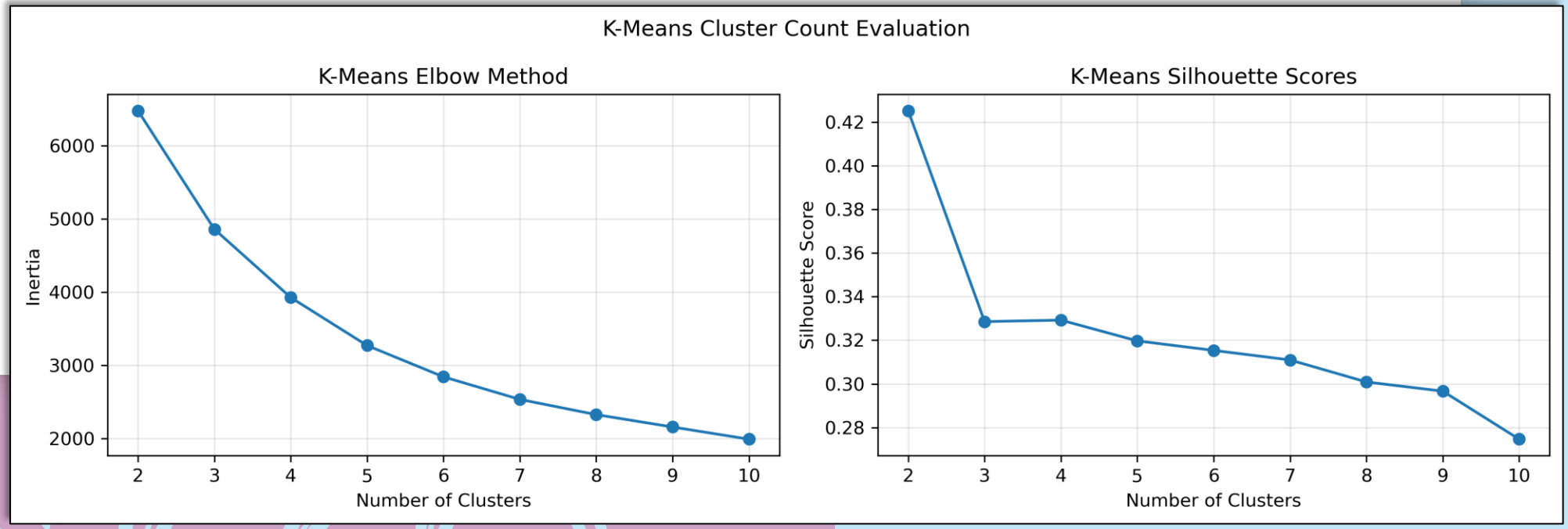
Preparing the Features for Clustering

- Frequency and spending were strongly skewed
- Applied logarithmic transformations
- Standardized the three features
- Preserved original values for interpretation



Why Four Customer Segments

- Evaluated **2-10 clusters**
- Compared inertia and sampled silhouette scores
- Two Clusters: 0.4252
- Four Clusters: 0.3292
- Four clusters provided more useful customer detail



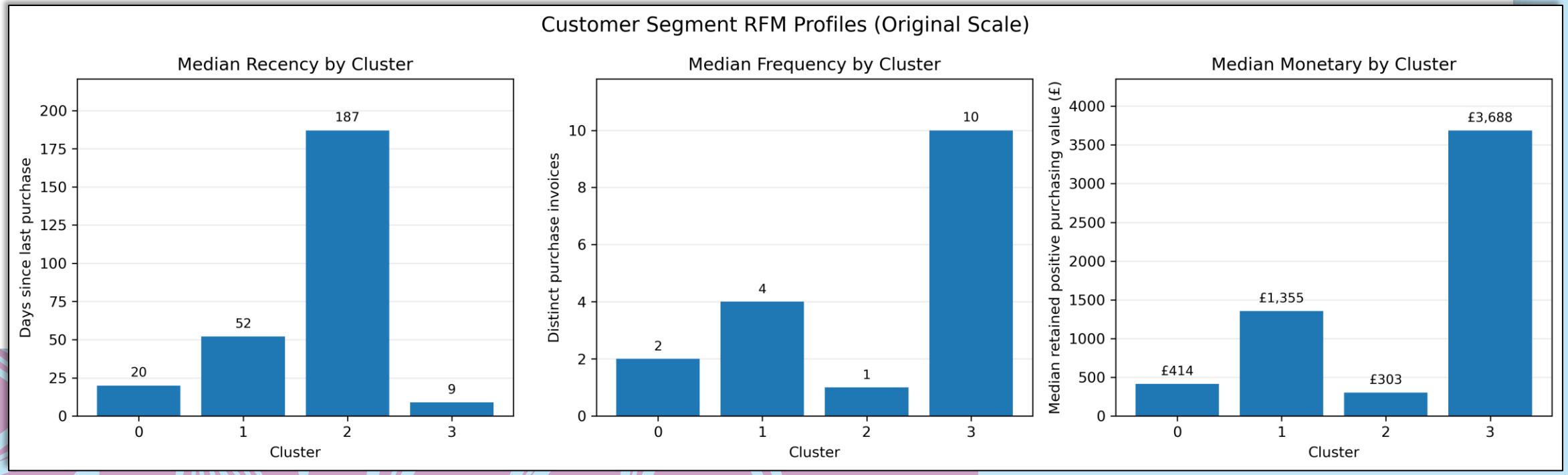
Rousseeuw (1987);
scikit-learn developers (n.d.)

The Four Customer Segments

Cluster	Segment	Customers
0	Recent Low-Spend	897
1	Established Moderate-Value	1,174
2	Inactive Low-Value	1,549
3	High-Value Active	718

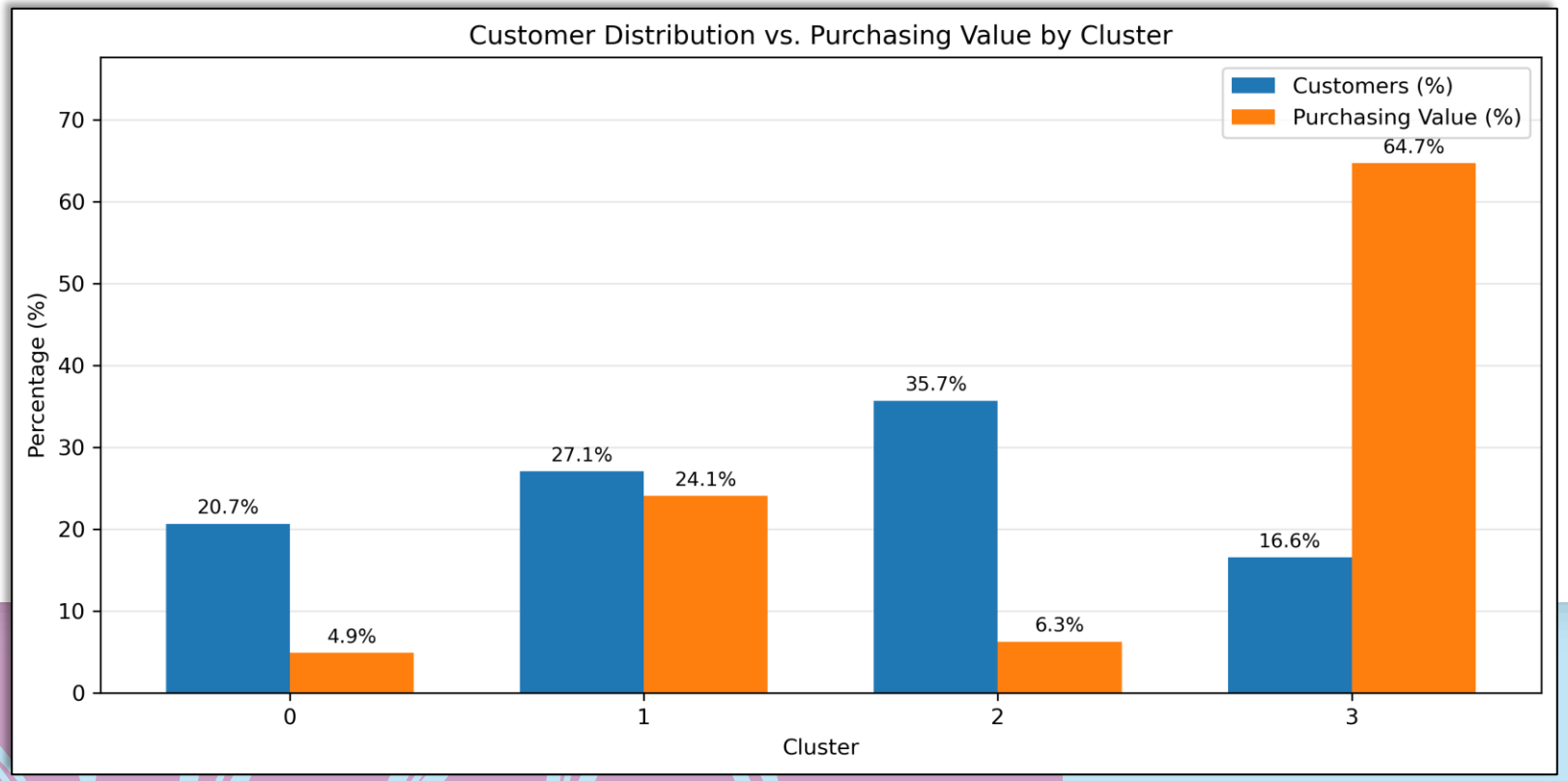
How Purchasing Patterns Differed

- High-value active customers: median **9 days** since purchase and **10 invoices**
- Inactive low-value customers: **median 187 days** and **1 invoice**
- Lower recency means a more recent purchase



Customer Share and Purchasing Value

- High-value active: **16.55% of customers -> 64.73% of purchasing value**
- Inactive low-value: **35.71% -> 6.28%**
- Purchasing value is **not profit or net revenue**



Actions Worth Testing

Segment	Potential Approach
Recent Low-Spend	Follow-up to encourage another purchase
Established Moderate-Value	Relevant repeat-purchase offers
Inactive Low-Value	Selective re-engagement
High-Value Active	Retention, service, and loyalty efforts

- Compare campaigns with control groups

Limitations and Responsible Use

- Historical data from one retailer
- Exclusions affect customer representation
- Three features capture only part of customer behavior
- Segment stability and campaign effectiveness remain untested
- Protect purchasing records and review targeting decisions

Implementation and Main Takeaway

- Validate current retailer data
- Calculate RFM and fit preprocessing and clustering
- Evaluate profiles and stability
- Test selected strategies
- Monitor outcomes and refresh the analysis
- **Takeaway**: Four descriptive segments provide useful questions for customer engagement

References

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- Hughes, A. M. (1994). *Strategic database marketing*. Probus Publishing.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics, 20*, 53–65. https://doi.org/10.1016/0377-0427(87)90125-7
- Scikit-learn developers. (n.d.). *KMeans*. Retrieved from https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html

Project Repository: https://github.com/ferrillt/Customer-Segmentation-Clustering